SAND & DUST STORM (SDS) FORECASTING TAXONOMY SITEMAP

Comparative Hierarchy: Traditional Methods vs. Multi Scale vs. Machine Learning vs. Deep Learning (LSTM/Transformer/GNN) vs. Tri Pillar

BRANCH 1: TRADITIONAL METHODS

Synoptic, Classical Empirical & NWP/CTM

1.1 Subjective Synoptic Analysis

- Manual surface isobaric chart tracing
- 500 hPa troughs & 850 hPa wind convergence
- Delta P > 3-4 hPa pressure gradient rules
- Subjective forecaster experience heuristic

1.2 Classical Empirical Saltation

- Persistence & 30-year climatology baseline
- Linear regressions & ARIMA time series
- Bagnold (1941) sand flux: q = C(u_*)_3
- Owen (1964) saltation threshold gate: u_* > u_*t

1.3 Dynamical NWP & CTM Models

- Global NWP: ECMWF IFS, CMA GFS
- Atmospheric CTM: WRF-Chem, DREAM8b, CAMS
- Hydrodynamic Navier-Stokes equations
- Aerosol continuity: dC/dt + div(vC) = K div^2 C + S - vd C

1.4 Operational Bottlenecks [CRITICAL]

- Chaotic Lyapunov Limit: collapses after Day 3
- Coarse 9-25km grid misses Wadi Hanifa escarpments
- Inability to model convective Haboob downburst walls
- Extreme supercomputer HPC burden (hours per cycle)

BRANCH 2: MULTI-SCALE DYNAMICS

Small, Medium & Large Space/Time Scales

2.1 Small Scale (Microscale & Short-Term)

- Spatial: < 2 km (microscale boundary layer shear)
- Temporal: 0-2 hr (nowcasting & acute alerts at KKIA)
- Hazard Level 1-2 (Mild haze; PM10 < 150 ug/m3)
- Friction velocity u_* & local roughness z_0 gating

2.2 Medium Scale (Mesoscale & Extended)

- Spatial: 2-2,000 km (convective Haboob dust walls)
- Temporal: 3-15 days (Sub-seasonal) - Core Focus!
- Hazard Level 3-4 (Blowing sand; 150-2,000 ug/m3)
- Municipal pre-wetting & flight rerouting window

2.3 Large Scale (Synoptic & Seasonal)

- Spatial: > 2,000 km (regional Shamal regimes)
- Temporal: > 15-90+ days (seasonal drought memory)
- Hazard Level 5 (Catastrophic blanket Haboob > 2,000)
- Teleconnections: ENSO (ON/MJO), NAO, AO, IOD

2.4 5-Tier Hazard Classification

- Level 1: Normal air (PM10 < 50 ug/m3, Vis > 10km)
- Level 2: Slight dust (PM10 50-150 ug/m3)
- Level 3: Moderate blowing sand (PM10 150-500 ug/m3)
- Level 4-5: Severe Haboob (PM10 > 2,000 ug/m3, Vis < 200m)

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BRANCH 3: MACHINE LEARNING (LINE A)

Statistical ML, GBDT Ensembles & Post-Processing

3.1 Classical ML (Literature Cited)

- Artificial Neural Networks (ANN / Multi-Layer Perceptrons)
- Support Vector Machines (SVM) with 500 hPa fields
- Fuzzy logic (ANFIS) & Radial Basis Functions (RBF)
- Bayesian Networks + Case-Based Reasoning (CBR)

3.2 GBDT Tree Ensembles

- LightGBM: ultra-fast leaf-wise histogram splits
- CatBoost: ordered target encoding for station data
- XGBoost: regularized depth-wise gradient boosting
- AutoML: AutoGluon & Auto-TPOT pipelines

3.3 Imbalance & Cost-Sensitive Learning

- The < 3% rare storm frequency challenge
- SMOTE minority oversampling for extreme events
- Asymmetric cost matrix (false negative penalty 10x)
- Focal Loss down-weighting easy normal weather

3.4 NWP Bias Correction & Quantiles

- Residual learning: y_true ~ NWP + f_GBDT(X_env)
- Pinball loss quantiles: P10, P50 (median), P90 (risk)
- Calibrated non-crossing prediction intervals
- Tree SHAP game-theoretic feature attribution

BRANCH 4: DEEP LEARNING (LINE B)

LSTM, Transformer & GNN Architectures

4.1 Long Short-Term Memory (LSTM)

- Recurrent temporal gating for sequential accumulation
- Bidirectional LSTM/GRU with temporal attention
- Captures multi-step ahead continuous concentrations
- Solves long-range vanishing temporal gradients

4.2 Transformer & Attention Networks

- Multi-head self-attention for sub-seasonal waves
- Spatiotemporal attention capturing planetary forcing
- Models non-local teleconnection tele-couplings
- Maintains robust predictive attention into Day 15

4.3 Corridor Graph Networks (ST-GNN)

- 14-Node directional transport corridor topology
- Asymmetric Chebyshev polynomial graph convolutions
- Captures directional upwind-to-downwind transport
- Models Ad-Dahna & An-Nafud outflow into Riyadh

4.4 Computer Vision & Foundation AI

- 2D/3D CNNs extracting vortex circulations from grids
- Transfer learning on FengYun FY-4 & MODIS AOD
- Coupled AI-GAMFS 5km grid (twice-daily cycles)
- 38%-74% error reduction vs standard ECMWF/NASA

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BRANCH 5: WHAT SHE USES (TRI-PILLAR)

AI (LSTM/Transformer/GNN) + GIS + SEM/PADM

5.1 Pillar 1: AI (LSTM + Transformer + GNN)

- Dual-line: Line A GBDT + Line B LSTM/Transformer/GNN
- Replaces NWP with pure data-driven deep triad
- Extends 3-15 days forecast with P10/P50/P90 envelopes
- Tree SHAP + attention rollouts for XAI diagnostics

5.2 Pillar 2: Spatial Geography & GIS

- Ordinary Kriging spatial exposure & vulnerability maps
- Global Moran's I & LISA local high-high cluster maps
- Geographically Weighted Regression (GWR) on z_0
- 90m SRTM DEM topographical escarpment analysis

5.3 Pillar 3: Socioeconomic Behavioral Science

- IBM SPSS: Little's MCAR missing test & EFA screening
- IBM AMOS: Confirmatory Factor Analysis (CFA)
- Construct Reliability CR >= 0.70 & Convergent AVE >= 0.50
- Protective Action Decision Model (PADM); SEM compliance

5.4 Municipal Crisis Early Warning Playbook

- 4-Tier operational matrix: T-15d -> T-5d -> T-24h -> T-0h
- Aviation: flight holds & runway diversions at KKIA
- Urban: high-pressure street misting & cannon deployment
- Health: pre-emptive respiratory medication stocking

METHODOLOGICAL PARADIGM EVOLUTION (SITEMAP SUMMARY)

Traditional NWP (Chaotic 72h Lyapunov Barrier) → Line A GBDT Bias Correction (Quantiles P10/P50/P90) → Line B Deep AI (LSTM + Transformer + Corridor GNN) → Tri-Pillar Delivery (GIS Kriging + SPSS/AMOS PADM)